

Practical 13 - Alternative models for survival data

Survival Analysis

London School of Hygiene & Tropical Medicine

Stata version (please see accompanying Stata do file)

Datasets required: `pbcbase.dta`

Introduction

In this practical we will return to using the `pbcbase` data, used in earlier practicals. The data are from a randomised trial and the time origin is the time of randomisation. The variables of interest for this practical are described below.

Variable	Description
<code>id</code>	Unique identifier for each participant
<code>d</code>	Event indicator at the end of follow-up: 0=alive (censored), 1=dead
<code>time</code>	Follow-up time in years
<code>treat</code>	Treatment: 1=placebo, 2=active
<code>bil0</code>	Serum bilirubin (mg/dl), measured at the start of the trial.

The practical is in 3 parts:

- Part A: Flexible parametric models
- Part B: Accelerated failure time models
- Part C: Additive hazards models

Aims

The aims of this session are:

- Fit flexible parametric models (FPM) using a spline for the baseline hazard, and obtain corresponding survival curves.
- Fit Weibull models and make the link between proportional hazards and accelerated failure time (AFT) parametrizations.
- Fit Weibull and log-logistic AFT models and compare the fit.
- Fit additive hazard models and interpret the graphical output.

Part A: Flexible parametric models

Question 1

In this question we will fit a flexible parametric model (FPM) to the PBC data, with the baseline hazard modelled using a spline (as a function of time).

Use the code provided to carry out the following steps: - Load the data and apply `stset`. - Start by calculating the 25th, 50th and 75th percentiles (i.e. the quartiles) of follow-up time for individuals who have the event. - Using `strcs` fit a FPM including `treat` and `bil0` as covariates. For the baseline hazard use a natural cubic spline with 3 interior knots for $\log(t)$ located at the quartiles of log follow-up time in people who have the event. - Take a look at what the output is showing and link this to what was shown in the slides.

Question 2

- Use the code provided to obtain estimated survival curves based on the FPM fitted in question 1 for two individuals: person 1 has `treat=1`, `bil0=30` and person 2 has `treat=2`, `bil0=30`.
- Fit a new FPM with 10 internal knots instead of 3. Compare the coefficients from the two models.
- Compare estimated survival curves from the two models for the two individuals above.

Question 3

- Compare your hazard ratio estimates from the FPM (with 3 knots) with those from using a Cox proportional hazards model.
- Compare the estimated survival curves for the two individuals considered above.
- What are the assumptions of the FPM versus the Cox model?

Part B: Accelerated failure time models

Question 1

- Fit a Weibull model with treatment group as the explanatory variable, using the proportional hazards parametrization. This can be done using `streg` with the option `distribution(weibull)`.
- Write down the model being fitted and interpret your results.

Question 2

- Fit a Weibull model with treatment group as the explanatory variable, using the AFT parametrization. This can be done using the `time` option at the end of `streg`.
- Write down the form of the survival function under the AFT Weibull model and interpret your results.
- What is the estimate of the acceleration factor $e^{\beta_{AFT}}$? Interpret the estimate.
- What is the relationship between your estimates of the hazard ratio $e^{\beta_{PH}}$ and the acceleration factor $e^{\beta_{AFT}}$?

Question 3

- A log-logistic model is another type of AFT model. Use `streg` to fit a log-logistic model with treatment as the explanatory variable, using the option `distribution(loglogistic)`.
- Find the estimates of the acceleration factor, and interpret the results.
- Compare the fit of the Weibull model and the log-logistic model using the AIC. Which model provides the better fit?

Part C: Additive hazard models

Question 1

- Fit the additive hazards model using `treat` and `bil0` as explanatory variables.
- Write down the form of the model fitted and take a look at the estimated coefficients.
- Plot the cumulative coefficients and interpret the graphical output.

Question 2

In this question we will compare survival curves estimates using the additive hazards model and using the Cox model.

The code below takes you through the following steps:

- We first need to use a different function for fitting the additive hazards model (`aalen` in the `timereg` package) because using `aareg` in the `survival` package does not allow us to automatically obtain survival estimates.
- Based on the additive hazards model fitted above, obtain the estimated survival curves for individuals 1 and 2: person 1 has `treat=1`, `bil0=30` and person 2 has `treat=2`, `bil0=30`.
- Compare the estimated survival curves with those obtained from the Cox model in Part A.